

SUMIN PARK

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 [Linkedin](#) |  [Github](#) |

EDUCATION

- **Korea Advanced Institute of Science and Technology (KAIST)** Sep 2024 – Present
M.S. in Computer Science Daejeon, South Korea
 - GPA: 4.18 / 4.30
- **Ewha Womans University** Mar 2016 – Feb 2021
B.S. in Brain and Cognitive Sciences & B.S. in Life Sciences Seoul, South Korea
 - GPA: 4.07 / 4.30 (Summa Cum Laude)
 - Thesis: *Differential Mechanism of Positive and Negative Valence Representation: A Focus on Amygdalar Circuitry*

EXPERIENCE

- **Big Data Analysis & Learning Laboratory (BigDyL), KAIST** Sep 2024 – Present
Master's Student Researcher Daejeon, South Korea
 - Advisor: Noseong Park
 - Research Topic: Interpretable LLM architectures, State-Space-Models (SSMs), Mixture-of-Expert (MoE), Representation learning, Brain-inspired AI.
- **Big Data Analysis & Learning Laboratory (BigDyL), Yonsei University** Jul 2023 – Aug 2024
Research Intern Seoul, South Korea
 - Advisor: Noseong Park
 - Research Topic: Graph representation learning, Graph neural networks (GNN)
- **DNI Consulting** Feb 2022 – Apr 2024
Data Scientist Seoul, South Korea
 - Research Topic: Data analysis and modeling for B2B marketing and customer segmentation
- **Cell Physiology Laboratory, Seoul National University** Aug 2020 – Jan 2021
Undergraduate Research Intern Seoul, South Korea
 - Advisor: Seokho Lee
 - Research Topic: Electrophysiology, PFC-involved systems consolidation
- **Neurobiology Laboratory, Seoul National University** Dec 2019 – Feb 2020
Undergraduate Research Intern Seoul, South Korea
 - Advisor: Bongkiun Kaang
 - Research Topic: Molecular analysis on dopaminergic reward circuitry
- **Molecular Neuroscience Laboratory, POSTECH** July 2019 – Sep 2019
Undergraduate Research Intern Pohang, South Korea
 - Advisor: Junghun Kim
 - Research Topic: Extinction of cocaine-context association memory

PUBLICATIONS

INTERNATIONAL CONFERENCES

- [C6] **Q-Delta: Beyond Key-Value Associative State Evolution** | *Under Review*
 - [Sumin Park](#), Seojin Kim, and Noseong Park
- [C5] **STAR: Rethinking MoE Routing as Structure-Aware Subspace Learning** | *Under Review*
 - [Sumin Park](#) and Noseong Park
- [C4] **How Many Experts Are Enough? Towards Optimal Semantic Specialization for Mixture-of-Experts** | *AAAI 2026*
 - [Sumin Park](#) and Noseong Park

- [C3] **Are Graph Transformers Necessary? Efficient Long-Range Message Passing with Fractal Nodes in MPNNs** | AAAI 2026 (Oral)
- Jeongwhan Choi, Seungjun Park, **Sumin Park**, Sung-Bae Cho and Noseong Park
- [C2] **DARS: Robust Sparse Fine-Tuning with Regularized Subspace Disalignment** | ICLR 2025 Workshop (SCOPE)
- **Sumin Park** and Noseong Park
- [C1] **PANDA: Expanded Width-Aware Message Passing Beyond Rewiring** | ICML 2024
- Jeongwhan Choi, **Sumin Park**, Hyowon Wi, Sung-Bae Cho, and Noseong Park

INDUSTRIAL PROJECTS

- **Simulation-Based Defect Variable Identification from High-Resolution Images** Mar 2025 – Dec 2025
 - Ongoing **Samsung-KAIST** industry-academia project
 - Developed a multi-task, multi-class image classification framework to predict defective variables from simulation-generated images
 - Designed a ViT architecture with hierarchical attention for efficient high-resolution image processing, achieving 99.41% classification accuracy

LEADERSHIP AND EXTRACURRICULARS

- **Ewha Institute of Brain & Cognitive Science, Ewha Womans University** Spring 2020 – Fall 2020
President
 - Led and organized academic activities including weekly open research sessions and end-of-semester undergraduate journal publications
 - Reviewed multiple journals per session and authored weekly summary reports
- **Debate Association of Ewha (DAE), Ewha Womans University** Spring 2019 – Fall 2019
Regular Member
 - Participated in weekly team debate sessions
 - Competed in national debate tournaments (SRT, KNC); SRT 2019 Quarter Finalist

HONORS AND AWARDS

- Summa Cum Laude, Ewha Womans University 2016 – 2020
- Dean's List (7 semesters), Ewha Womans University 2016 – 2020
- Academic Honors Scholarship, Ewha Womans University Spring 2020
- Global Leader in Major, Ewha Womans University Spring 2020
- Academic Honors Scholarship, Ewha Womans University Spring 2019
- Admissions Scholarship (Full Tuition, 2 Years), Ewha Womans University 2016 – 2018

SKILLS

- **PROGRAMMING & FRAMEWORKS:** Python, R, C++, PyTorch, Triton
- **SYSTEMS & TOOLS:** Linux, Git, Docker, AWS, SQL